Cash Radar: spotting a cash shortage before it happens

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Cash Radar is a new feature for Financial Navigator that predicts a company's cash, week by week, for the next 13 weeks. It combines the money in the bank, the money customers are expected to pay and the bills the company must pay. If cash will fall too low, it warns the finance manager (CFO) weeks in advance, explains why in plain words and suggests actions. A person always makes the final decision.

1. Summary

Problem. A small or medium-sized company (SME) can make a profit and still run out of cash. Profit is counted when a sale is invoiced, but cash only arrives when the customer pays. If customers pay late and a large supplier bill is due, the bank account can empty while the accounts still show a profit. Many SMEs still use a hand-made spreadsheet that assumes customers pay on time.

Main idea. Financial Navigator is already connected to the company's bank accounts. Cash Radar adds invoices, orders and bills to build a weekly cash forecast. It expects each customer to pay with their usual delay, not on the due date. Each week shows a likely range, not one number. If any week falls below the minimum cash the CFO wants to keep (the "buffer"), Cash Radar shows a warning and possible actions.

Users. CFOs and finance teams of small and medium-sized companies, who are Financial Navigator's main customers, and the accountant who keeps the data up to date.

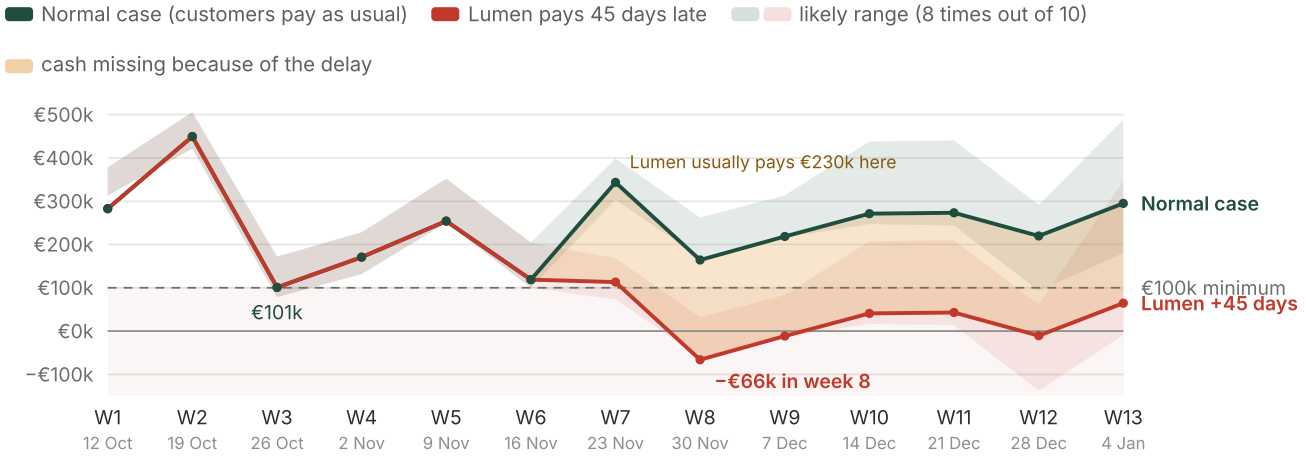
Expected benefits (assumptions, to be checked in a pilot):

- A warning at least 4 weeks before cash runs short.
- Forecasts at least 50% more accurate than a spreadsheet that assumes on-time payment.
- Less time spent preparing the cash forecast each week.
- Profit and cash shown side by side, so a good profit does not hide a cash problem.

> **Example (fictional company).** Atelier Verde is a furniture maker with 38 staff, €381k in the bank and a minimum cash target of €100k. It expects a **profit of €182k this quarter**. But if its biggest customer, Lumen Hospitality, pays a €230k invoice 45 days late, cash drops to **−€66k in the week of 30 November 2026**, just after a €155k timber bill. Cash Radar warns about this **7 weeks in advance**.

+€182k	−€66k	7 weeks	75%
expected profit this quarter	lowest cash if Lumen pays 45 days late (week 8)	warning time before cash falls too low	smaller forecast error than assuming on-time payment

Figure 1 · One late payment turns a tight quarter into an overdraft



Cash in the bank at the end of each week, 12 Oct 2026 – 10 Jan 2027, in euros. Fictional data. The range comes from 4,000 simulations of when customers might pay.

2. AI approach

Cash Radar uses **simple statistics that are easy to explain, and basic calculation everywhere else**. Generative AI is used only to write the warning message, using numbers that are already calculated. So every number can be traced back to a real invoice, bill or bank transaction.

Data sources

- **Bank accounts**: balances and transactions that Financial Navigator already receives.
- **Money coming in**: unpaid customer invoices, at least 12 months of paid invoices with their payment dates, and

How it works, every Monday

1. **Update.** Real bank balances replace last week's forecast, and the difference is recorded to measure accuracy.
2. **Predict customer payments.** Each invoice is expected on its due date plus that customer's usual delay over the last 12

confirmed orders.

- **Money going out**: supplier bills and orders, salaries, social charges, VAT, taxes, rent and loan repayments, from the accounting software.

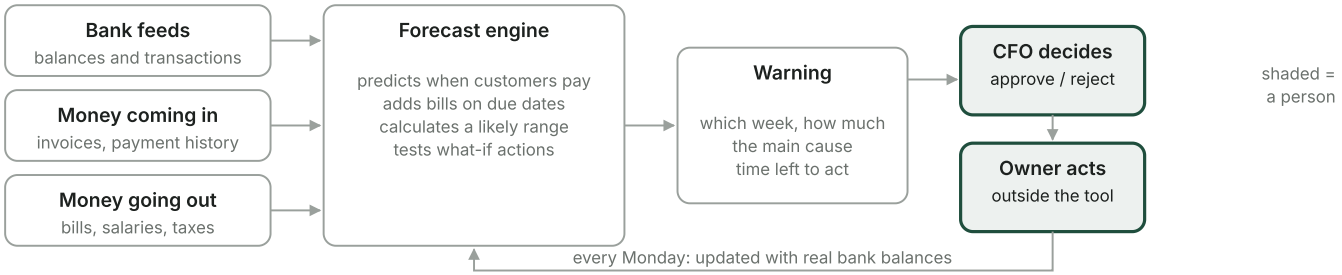
Where people stay involved

The accountant checks customer payment habits and settles disputed invoices. The CFO sets the minimum cash level, can change any assumption (each change is saved with a comment) and approves or rejects every action. Cash Radar never moves money, never contacts customers and never contacts the bank.

months. For example, Bâtir Bureaux usually pays about 21 days late, while Officea pays exactly on time.

3. **Add payments** on the dates they are due (salaries, taxes, bills).
4. **Show uncertainty.** The forecast is repeated 4,000 times with different past delays, giving a likely range and the chance of falling below the minimum.
5. **Test "what if" situations**, such as "Lumen pays 45 days late" or "split the timber bill into two".
6. **Warn** if any week falls below the minimum: which week, how much is missing, why, and how sure it is.

Figure 2 · The weekly cycle: Cash Radar suggests, a person decides and acts



Why not a complex AI model? A simple method is easier to understand and trust. On the fictional company's past data, using each customer's usual delay cut the average weekly error from **€304k** to **€77k**. A more complex version was only 14% better, so it is left for later.

Example warning (written from calculated numbers). "If Lumen pays 45 days late, cash will fall to −€66k in the week of 30 November, which is €166k below your €100k minimum. The cause is timing, not profit: the €230k payment arrives too late, just after the €155k timber bill. You have 7 weeks to act. If the delay happens, a shortfall is almost certain. This quarter's profit is still +€182k. Possible actions are listed below; nothing happens without your decision."

#	Possible action for the CFO (if Lumen pays 45 days late)	Improves the lowest cash by	Cost or downside	Who acts
A1	Ask the two slowest customers to pay on time	+€82k	Staff time; the contract allows late-payment fees	Accountant
A2	Re-send a hospital invoice that was rejected	+€17k	None	Accountant
A3	Delay a €34.6k machine deposit to January	+€35k	The machine arrives about 6 weeks later	CEO / CFO
A4	Ask the timber supplier to accept two payments	+€56k	Might lose part of a 3% discount (up to €4.7k)	CFO
A5 / A6	Offer Lumen 2% to pay on time / borrow against the invoice	+€167k	About €4.6k discount / about €3–4k in fees	CEO / CFO
A7	Use the €150k bank overdraft	only covers cash below €0	About 4.6% interest; the bank sees the company is under pressure	CFO

Actions A1–A4 together bring the lowest cash back to €100k. Adding A6 keeps cash above the minimum in every week. For each action the CFO can choose "Approve", "Reject" or "More information", and the choice is recorded.

3. Fairness

Reducing bias. Payment habits are learned only from each customer's own invoices. Nothing about the owner, industry, size or location is used. Customers with few invoices are given the company's average delay, not a harsh guess. Disputed invoices are left out until settled, so a complaint about quality is not counted as late payment. If a customer starts paying 7+ days slower, the tool suggests a conversation, never a penalty.

Clear results. Every warning says when, how much, why and how sure. Users can click any weekly number to see the invoices and bills behind it. Profit and cash are always shown together.

Accessibility. Colour is never the only signal; text and labels carry the meaning too. Charts have a text version, the screen works on a phone, and the weekly table exports to Excel.

Correcting errors. Users can change any payment date or assumption, with a comment. Every change is saved and can be undone. Each week, the forecast is compared with what really happened.

4. Plan

This proposal is about the idea; the plan below only shows how it could be tested. **A small pilot with a few existing Financial Navigator clients** would run Cash Radar next to their current forecast.



People involved. A352's product team (makes the final decision), a data analyst (builds and tests the forecast), a developer (connects the data and builds the screens), the CFO of each pilot client (judges whether warnings are useful) and A352's data protection officer.

Costs and resources. To be agreed with A352 if the idea is accepted. The pilot needs no new data or equipment: it uses the bank connections and client data Financial Navigator already has, a small team and standard cloud services.

Checks before launch. The feature is launched only if:

- it is clearly more accurate than the client's current forecast;
- it would have warned about past cash problems weeks in advance, without too many false alarms;
- privacy and security checks are approved;
- no action can happen without a person's approval;
- pilot CFOs find the warnings useful and easy to understand.

5. Impact

Indicator	How it is measured	Result on fictional data	Pilot target (assumption)
Forecast error	Average difference between the forecast and the real bank balance each week, compared with a forecast that assumes on-time payment	€77k vs €304k (75% smaller) over 66 tests; still €54k too optimistic on average	At least 50% smaller than the client's current method
Warning time	Number of weeks between the first warning and the week cash falls below the minimum. The CFO also rates each warning as useful or not	7 weeks now; one past cash drop flagged 12 weeks early, one missed (too little history)	4+ weeks before every real shortfall; fewer than 1 in 3 false alarms
Time saved	Hours per week spent preparing the cash forecast, recorded before and during the pilot	Cannot be measured with fictional data	Clearly less time (to be measured)

Accuracy and warning time are measured automatically every week. Financial benefits, such as lower interest costs, stay assumptions until the pilot measures them.

6. Risks

Area	Risk	How we reduce it
Privacy	Some customers are individuals, and salary data is personal, so EU data protection law (GDPR) applies	Use only invoice payment data; salaries as one monthly total; data kept in the EU and deleted after 24 months; a privacy review before the pilot
Security	Bank and invoice data are sensitive and could attract hackers	Use Financial Navigator's existing security and access rights; Cash Radar can read data but not change it; the AI text tool receives only totals, never raw data
Ethics	Unfairly labelling customers as "bad payers"; putting pressure on suppliers	Judge payment habits only; flags can be reviewed; supplier actions are polite requests within agreed terms
Legal	EU AI Act rules on being open about AI; French limits on payment terms (60 days, or 30–50 days for public buyers); giving financial advice	AI-written text is labelled; suggested actions respect legal payment terms; the tool only informs, the CFO decides; it does not sell loans
Forecast quality	Missing data, unusual events or customers changing habits; the forecast being too optimistic	Weekly check against real results; alerts when habits change; manual corrections; the pessimistic end of the range is always shown
Too much trust	Too many alerts are ignored, or one number is trusted blindly	The range is always shown; every alert is rated; no action happens automatically

7. Tools and sources

Fictional data. Atelier Verde and all its data are invented. Real French rules and Banque de France statistics were used only to make it realistic. Section 5 targets are assumptions.

Tools used. Claude (Anthropic), a generative AI assistant, for the dataset, code, text and dashboard mock-up · **Python** for the simulation, forecast and tests · **Excel** workbook · **"Cash Radar" dashboard** (mock-up, not a working product). The author checked all figures and is responsible for the content.

- Banque de France, Observatoire des délais de paiement, 2025 report and press release, 24 Sep 2026: average payment delays in France.
- Code de la commande publique: payment limits for public buyers.
- RubyPayeur, LME law: payment limits between companies, late-payment fees.
- Swim Legal, URSSAF and VAT (CA3); Bpifrance: tax and charge dates.
- Kognitos: 13-week forecast practice; Dealroom: A352 bank connections.